

Report

[SBA 928 - Enhancing Market Research with AI Prompt Engineering]

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[07/31/2026]



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Overview

This assessment focuses on applying advanced AI prompt engineering techniques to improve market research. The objective is to create diverse market research prompts, develop prompt variations, evaluate the effectiveness of fine-tuning language models, and identify potential biases that may affect AI-generated insights. By completing this assessment, learners gain practical experience in generating accurate, relevant, and fair market research outputs that support informed business decisions and strategic planning.

Steps

Part 1: Prompt Engineering

1. Selected Market Research Dataset

The **Marketing Campaign** dataset from Kaggle was selected for this assessment. It contains **2,240 customer records** and **29 variables**, including customer demographics, income, education, purchasing behavior, product spending, web and store purchases, campaign responses, and complaints. The dataset is suitable for analyzing **consumer behavior**, **market trends**, and **marketing campaign effectiveness**.

Screenshoot 1. Market Research Dataset

ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Customer	Recency	MktWines	MntFruits	MeatProducts	FishProducts	SweetProducts	MntGold	NumDealsPromo	WebPurchases	CatalogPurchases	StorePurchases	WebVisits	AcceptedCampa	AcceptedCampa	AcceptedCampa	AcceptedCampa	AcceptedCampa	Complaints	CostContact	Z_Revenue	Response
5524	1957	Graduation	Single	58138.0	0	0	04-09-2012	58	635	88	546	172	88	88	3	8	10	4	7	0	0	0	0	0	0	3	11	1
2174	1954	Graduation	Single	46344.0	1	1	10-03-2014	38	11	1	6	2	1	6	2	1	1	2	5	0	0	0	0	0	0	3	11	0
4141	1965	Graduation	Together	71613.0	0	0	02-08-2013	26	426	49	127	111	21	42	1	8	2	10	4	0	0	0	0	0	0	3	11	0
6182	1984	Graduation	Together	26646.0	1	0	02-02-2014	26	11	4	20	10	3	5	2	2	0	4	6	0	0	0	0	0	0	3	11	0
5324	1981	PHD	Married	58293.0	1	0	09-01-2014	94	173	43	118	46	27	15	5	5	3	6	5	0	0	0	0	0	0	3	11	0
7446	1967	Master	Together	62513.0	0	0	10-09-2013	16	520	42	98	0	42	14	2	6	4	10	6	0	0	0	0	0	0	3	11	0
965	1971	Graduation	Divorced	55635.0	0	0	13-11-2012	34	235	65	164	50	49	27	4	7	3	7	6	0	0	0	0	0	0	3	11	0
6177	1985	PHD	Married	33454.0	1	0	08-05-2013	32	76	10	56	3	1	23	2	4	0	4	8	0	0	0	0	0	0	3	11	0
4855	1974	PHD	Together	30351.0	1	0	06-06-2013	19	14	0	24	3	3	2	1	3	0	2	9	0	0	0	0	0	0	3	11	1
5899	1950	PHD	Together	5648.0	1	1	13-03-2014	68	28	0	6	1	1	13	1	1	0	0	20	1	0	0	0	0	0	3	11	0

2. Developed Prompts

Prompt 1 – Consumer Behavior

Analyze the Marketing Campaign dataset and identify customer purchasing behavior by age, income, education, and product category. Summarize the main spending patterns and recommend strategies to improve customer engagement.

Screenshoot 2. Prompt 1 & Output – Consumer Behavior

Analyze the Marketing Campaign dataset and identify customer purchasing behavior by age, income, education, and product category. Summarize the main spending patterns and recommend strategies to improve customer engagement.

Worked for 17s >

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Customer Purchasing Behavior Analysis – Marketing Campaign Dataset

Executive Summary

The **Marketing Campaign** dataset was analyzed to evaluate customer purchasing behavior based on **age, income, education, and product category**. The analysis shows that **income is the strongest predictor of customer spending**, while customers with higher education levels also tend to spend more. Among all product categories, **wine** and **meat products** account for the largest share of customer purchases.

Customer Purchasing Behavior

By Age

Age Group	Average Spending
18–35	927
36–50	514
51–65	584
66+	717

Key Insight: Younger customers (18–35) and older customers (66+) showed the highest average spending, while customers aged 36–50 spent the least on average.

By Income

Income Group	Average Spending
High Income	1,058
Low Income	156

Key Insight: Customers with high incomes spend substantially more across all product categories, making them an important

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Prompt 2 – Market Trends

Analyze the Marketing Campaign dataset to identify trends in customer purchasing behavior, preferred shopping channels, and product categories. Highlight emerging trends and opportunities for future marketing campaigns.

Screenshoot 3. Prompt 2 & Output – Market Trends

Analyze the Marketing Campaign dataset to identify trends in customer purchasing behavior, preferred shopping channels, and product categories. Highlight emerging trends and opportunities for future marketing campaigns.

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Market Trends Analysis – Marketing Campaign Dataset

Executive Summary

The Marketing Campaign dataset was analyzed to identify trends in customer purchasing behavior, preferred shopping channels, and product categories. The analysis indicates that customers prefer **in-store purchases**, followed by **online purchases**, while **catalog purchases** are the least common. Wine and meat products account for the highest average spending, and customer income is strongly associated with total spending.

Customer Purchasing Behavior

Key Trends

- Higher-income customers consistently spend more across all product categories.
- Wine and meat products generate the highest customer spending.
- Customers generally purchase multiple product categories rather than a single product.
- The overall campaign response rate is relatively low (approximately 15%), indicating opportunities to improve marketing effectiveness.

Preferred Shopping Channels

Shopping Channel	Average Purchases
Store Purchases	5.79
Web Purchases	4.08
Catalog Purchases	2.66

Key Insight: Customers prefer shopping in physical stores, while online purchasing is the second most popular channel. Catalog purchases are used less frequently.

Product Category Trends

Product Category	Average Spending
Wine	303.94
Meat Products	166.95
Gold Products	44.02
Fish Products	37.53
Sweet Products	27.06
Fruits	26.30

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Prompt 3 – Competitor Analysis

Using the Marketing Campaign dataset, identify areas where the company can improve its marketing performance compared to competitors. Evaluate campaign effectiveness, customer response rates, and recommend competitive strategies.

Screenshoot 4. Prompt 3 & Output – Market Trends

Using the Marketing Campaign dataset, identify areas where the company can improve its marketing performance compared to competitors. Evaluate campaign effectiveness, customer response rates, and recommend competitive strategies.

Worked for 14s >

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Marketing Performance and Campaign Effectiveness Analysis

Executive Summary

The Marketing Campaign dataset was analyzed to evaluate campaign effectiveness, customer response rates, and opportunities to improve marketing performance. The dataset does not contain direct competitor information, so a true competitor comparison cannot be completed. However, internal performance gaps can be identified and used to recommend strategies that may strengthen the company's competitive position.

The latest campaign achieved a **14.9% response rate**, while previous campaign acceptance rates ranged from **1.3% to 7.5%**. Customers who responded to the latest campaign had higher incomes, spent more, purchased more frequently, and had made purchases more recently than nonresponders.

Campaign Effectiveness

Campaign	Acceptance Rate
Campaign 1	6.4%
Campaign 2	1.3%
Campaign 3	7.3%
Campaign 4	7.5%
Campaign 5	7.3%
Latest Campaign	14.9%

The latest campaign performed better than all five previous campaigns. However, approximately **85.1% of customers did not respond**, showing that targeting and personalization can still be improved.

Customer Response Analysis

Customer Measure	Responders	Nonresponders
Average income	\$60,210	\$50,839
Average total spending	\$987	\$539
Average purchases	15.37	12.04
Average recency	35 days	52 days
Previously accepted a campaign	56%	14%

Key Findings

Responders spent approximately **83% more** than nonresponders and completed more purchases. They had also purchased more recently and were much more likely to have

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More Detailed Prompt

Analyze the Marketing Campaign dataset by customer demographics, income, education, purchasing channels, and product categories. Identify purchasing trends and recommend marketing improvements.

Screenshoot 6. More Detailed Prompt & Output

Analyze the Marketing Campaign dataset by customer demographics, income, education, purchasing channels, and product categories. Identify purchasing trends and recommend marketing improvements.

Edit

Customer Purchasing Trends Analysis – Marketing Campaign Dataset

Executive Summary

The Marketing Campaign dataset was analyzed based on customer demographics, income, education, purchasing channels, and product categories. The analysis indicates that **income is the strongest factor influencing customer spending**, while education and purchasing behavior also contribute to customer value. Customers prefer shopping in physical stores, and wine and meat products represent the highest-selling categories.

Customer Demographics

Demographic	Key Finding
Age	Middle-aged and older customers account for most purchases.
Income	Higher-income customers spend significantly more than lower-income customers.
Education	Customers with higher education levels generally have higher spending.
Marital Status	Married and partnered customers represent a large portion of the customer base.

Purchasing Channels

Channel	Trend
Store Purchases	Most frequently used purchasing channel
Web Purchases	Second most popular channel
Catalog Purchases	Least frequently used channel

Key Insight: Although customers prefer in-store shopping, online purchases remain an important sales channel with opportunities for future growth.

Product Categories

Product Category	Customer Interest
Wine	Highest spending category
Meat Products	Second highest spending
Gold Products	Moderate spending
Fish Products	Moderate spending
Sweet Products	Lower spending
Fruits	Lowest spending

Purchasing Trends

Expert-Level Prompt

Generate a presentation-ready executive market research report using the Marketing Campaign dataset. Include customer segmentation, purchasing behavior, campaign effectiveness, market trends, KPIs, strategic recommendations, tables, and charts for senior management.

Screenshoot 7. Expert-Level Prompt & Output

Generate a presentation-ready executive market research report using the Marketing Campaign dataset. Include customer segmentation, purchasing behavior, campaign effectiveness, market trends, KPIs, strategic recommendations, tables, and charts for senior management.

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Executive Market Research Report

Marketing Campaign Dataset

Executive Summary

This report analyzes the **Marketing Campaign** dataset to evaluate customer segmentation, purchasing behavior, campaign effectiveness, and market trends. The objective is to provide senior management with actionable insights that support strategic decision-making and improve future marketing performance.

The analysis indicates that **high-income customers generate the highest revenue**, **wine and meat products are the strongest-selling categories**, **physical stores remain the preferred purchasing channel**, and the latest marketing campaign achieved the **highest response rate (14.9%)**, demonstrating improved campaign effectiveness.

Customer Segmentation

Segment	Key Insight
High-Income Customers	Highest overall spending and strongest campaign response
Middle-Income Customers	Largest customer segment with moderate spending
Lower-Income Customers	Lower purchasing frequency and spending
Higher Education	Greater average spending and customer value
Recent Purchasers	More likely to respond to marketing campaigns

Purchasing Behavior

Category	Finding
Highest Spending Products	Wine and Meat Products
Preferred Shopping Channel	Physical Store
Second Preferred Channel	Web Purchases
Least Used Channel	Catalog Purchases
High-Value Customers	Higher income and recent purchases

Campaign Effectiveness

Campaign	Response Rate
Campaign 1	6.4%
Campaign 2	14.9%

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4. Comparison of Responses

The prompt variations were tested using the same dataset. The **basic prompt** generated a simple summary of customer purchasing behavior. The **more detailed prompt** produced deeper insights by identifying customer segments, purchasing trends, and marketing opportunities. The **expert-level prompt** generated a structured executive report with an executive summary, KPIs, strategic recommendations, and presentation-ready formatting.

5. Effect of Prompt Changes

Increasing the level of detail, adding business context, and specifying the desired output format significantly improved the quality of the AI-generated responses. The responses became more accurate, organized, and relevant to business decision-making.

6. Most Effective Prompt

The **expert-level prompt** produced the most valuable market research insights because it included clear business objectives, detailed analysis requirements, structured formatting, and executive-level recommendations. This version generated the most comprehensive and actionable report for supporting strategic business decisions.

Part 2: Model Fine-Tuning

1. Structured Training Example

(Link: [flan_market_research](#) for exercise done in VSCODE)

Instruction (Prompt)

Analyze the Marketing Campaign dataset and identify customer purchasing behavior by income, education, product category, and shopping channel. Summarize the main trends and recommend marketing strategies to improve customer engagement.

Input (Context)

- Dataset: **Marketing Campaign (Kaggle)**
- Customer demographics (age, education, marital status)
- Income
- Product spending (Wine, Fruits, Meat, Fish, Sweets, Gold)

- Purchase channels (Web, Catalog, Store)
- Marketing campaign responses
- Customer complaints and recency

Expected Response

- Identify high-income customers as the highest-value segment.
- Identify wine and meat products as the top-selling categories.
- Recognize store purchases as the preferred shopping channel.
- Summarize purchasing trends across customer segments.
- Recommend personalized marketing campaigns, customer segmentation, loyalty programs, and digital marketing improvements.

2. Fine-Tuning Dataset Split

Model Selected: Google FLAN-T5-small (`google/flan-t5-small`)

Training-Data Format: JSONL containing three fields: instruction, context, and expected response.

Preprocessing: The Marketing Campaign dataset originally contained 2,240 customer records. Records with missing income values were removed, leaving 2,216 usable examples. Customer demographic information, income, recency, product spending, purchasing channels, and campaign response were converted into structured instruction-context-response examples.

Training / Validation / Evaluation Split:

Training Set: 1,772 examples (80%)

Validation Set: 222 examples (10%)

Held-Out Evaluation Set: 222 examples (10%)

```
PS C:\Users\arban\Downloads\flan_market_research> cd flan_sba928
PS C:\Users\arban\Downloads\flan_market_research\flan_sba928> uv run python prepare_data.py --csv marketing_campaign.csv
Clean rows: 2216
Training examples: 1772
Validation examples: 222
Held-out evaluation examples: 222
Spending thresholds: Q1=69.00, Q3=1048.00
PS C:\Users\arban\Downloads\flan_market_research\flan_sba928> █
```

Tokenization: FLAN-T5 tokenizer was used to convert the instruction and context into model input tokens and the expected response into target tokens.

Number of Epochs: 3

Batch Size: 2

Learning Rate: 5e-5

Maximum Input Length: 384 tokens

Maximum Response Length: 160 tokens

Adjustments During Training: Training examples were standardized to produce consistent market research outputs identifying customer value, product preferences, shopping-channel preferences, campaign response, and actionable recommendations. Instructions were also designed to discourage unsupported demographic assumptions.

Saved Model Checkpoint: `flan-market-research-final`

3. Base Model vs. Fine-Tuned Model Comparison

Evaluation Criteria	Base Model	Fine-Tuned Model
Relevance	Generated generic or unrelated responses.	Produced responses directly related to customer value, purchasing behavior, product categories, and marketing recommendations.
Specificity	Responses lacked detail and often included incomplete information.	Generated structured, detailed analyses including customer value, top product, preferred shopping channel, campaign response, and recommendations.
Factual Consistency	Frequently produced unsupported or inaccurate statements.	Responses were generally consistent with the Marketing Campaign dataset, although a few spending values differed from the expected outputs.
Completeness	Usually answered only one part of the prompt.	Addressed all required sections of the prompt in a consistent format.
Format Adherence	Responses did not follow the requested report structure.	Consistently followed the required instruction-response format.
Usefulness for Market Research	Limited practical value for business decision-making.	Produced actionable market research insights and marketing recommendations.
Potential Bias	Occasionally generated irrelevant demographic information without context.	Reduced unnecessary demographic assumptions and focused on customer purchasing behavior.

Screenshot 8. Base Model vs. Fine-Tuned Model Comparison

- More relevant responses to market research prompts.
- Better identification of customer purchasing behavior.
- Consistent report structure.
- Actionable marketing recommendations.
- Improved adherence to prompt instructions.
- Reduced irrelevant or repetitive responses.

Remaining Limitations

- Some customer spending totals differed from the expected values.
- A few product categories were predicted incorrectly.
- Customer value classifications were occasionally underestimated or overestimated.
- Additional training examples and more epochs could further improve accuracy.

4. Results

The fine-tuned FLAN-T5-small model demonstrated a clear improvement over the original base model. While the base model produced mostly generic responses, the fine-tuned model generated structured, business-focused market research analyses aligned with the Marketing Campaign dataset. These improvements make the fine-tuned model more suitable for customer segmentation, purchasing behavior analysis, campaign evaluation, and executive decision-making.

Bias and Fairness Analysis

Bias Assessment

The prompt-engineering process, fine-tuning dataset, and model outputs were evaluated to identify potential sources of bias that could affect the quality, fairness, and reliability of the AI-generated market research insights.

Demographic Representation

The Marketing Campaign dataset includes customer characteristics such as income, education, age, and marital status. However, some demographic groups may be underrepresented, which can influence the model's ability to generate equally accurate insights for all customer segments.

Regional Differences

The dataset is based on a single marketing campaign and does not include customers from multiple countries or geographic regions. Consequently, the model may not generalize well to customers with different regional purchasing habits or cultural preferences.

Sampling Bias

The fine-tuning dataset was created from one Marketing Campaign dataset. Since it represents a limited customer population, the model may learn patterns that are specific to that dataset rather than reflecting broader market behavior.

Industry-Specific Assumptions

The model was fine-tuned exclusively on retail marketing data. Therefore, its recommendations are appropriate for customer purchasing behavior and marketing analysis but may not be suitable for industries such as healthcare, banking, or manufacturing without additional training.

Bias in Expected Responses

The structured training examples consistently asked the model to identify customer value, top product, preferred shopping channel, campaign response, and a marketing recommendation. While this improved response consistency, it also encouraged the model to generate similar recommendation patterns across different customers.

Evaluation of Generated Outputs

The comparison between the base model and the fine-tuned model showed that the fine-tuned model generated more relevant, structured, and business-focused responses. However, the evaluation also revealed that the model sometimes predicted incorrect customer spending values, customer-value classifications, or top products. These errors indicate that although the model learned the expected response format, it did not always reproduce the exact numerical information from the dataset.

Fairness Recommendations

Issue Identified	Recommendation
Demographic representation	Expand the training dataset to include a more balanced distribution of customer ages, income levels, education backgrounds, and marital-status groups.
Regional differences	Include customer data from multiple regions and countries to improve the model's ability to generalize across different markets.
Sampling bias	Combine several marketing datasets rather than relying on a single campaign to reduce overfitting to one customer population.
Industry-specific assumptions	Develop separate fine-tuned models or training datasets for different industries instead of applying one model to all business domains.
Bias in expected responses	Increase the diversity of training examples and expected responses so the model learns multiple valid recommendation strategies rather than repeating similar outputs.
Numerical accuracy	Validate AI-generated outputs against the original dataset before using them for business decisions, especially when numerical values influence recommendations.
Human oversight	Include human review for important business analyses to verify customer classifications, spending values, and strategic recommendations before implementation.

The fine-tuned FLAN-T5-small model demonstrated a significant improvement over the original base model by producing more relevant, structured, and actionable market research insights. However, the evaluation also showed that the model occasionally generated incorrect numerical values and customer classifications. Expanding the training dataset, improving demographic diversity, validating outputs against the source data, and incorporating human review would further improve fairness, reliability, and the overall quality of AI-generated market research results.

Conclusion

This assessment demonstrated how AI prompt engineering and fine-tuning can improve market research. By refining prompts and fine-tuning the FLAN-T5-small model with the Marketing Campaign dataset, the quality of AI-generated insights improved significantly. The fine-tuned model produced more relevant, structured, and actionable responses than the base model. Although some numerical values were not always exact, the results showed that fine-tuning enhanced overall performance. Bias analysis also highlighted the importance of using diverse data, validating outputs, and incorporating human review to ensure fair and reliable business insights.

Appendix

Screenshoot 1. Market Research Dataset

Screenshoot 2. Prompt 1 & Output – Consumer Behavior

Screenshoot 3. Prompt 2 & Output – Market Trends

Screenshoot 4. Prompt 3 & Output – Market Trends

Screenshoot 5. Basic Prompt & Output

Screenshoot 6. More Detailed Prompt & Output

Screenshoot 7. Expert-Level Prompt & Output

Screenshoot 8. Base Model vs. Fine-Tuned Model Comparison